Sheet 8

Relaxation and cavity attenuation

1. Solve the lecture's Bloch equations from the excited state and from $(|0\rangle + e^{i\pi/3}|1\rangle)/\sqrt{2}$. Express the excited population and the $+x$ reading probability in terms of T_1, T_2, ν_0. Which measurement separates the two times?
2. A unital map contracts the Bloch components by (b, b, a), with $a = 0.6$, $b = 0.9$. Check positivity on isolated states, find the smallest Choi eigenvalue and its eigenvector, and the negative expectation after acting on half of $\Omega/\sqrt{2}$. Is it a channel?
3. **Vacuum cavity attenuation (longer).** A two-mode isometry is specified on number states by

$$J_\eta |n\rangle = \sum_{\ell=0}^{n} \sqrt{\binom{n}{\ell}} \eta^{(n-\ell)/2} (1-\eta)^{\ell/2} |n-\ell\rangle \otimes |\ell\rangle\,, \qquad 0 \le \eta \le 1.$$

   Use the endpoint convention $0^0 = 1$. Check its orthonormal images, then apply it to the coherent-state series. Find the reduced output and its mean number. With $\eta = e^{-\kappa t}$ and free phase rotation at angular frequency ν, find $\alpha(t)$. This computes the explicit attenuation channel; no general theorem about an unbounded Lindblad generator is requested.

> **First hint**
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> Two images of different number states belong to orthogonal total-number subspaces. For the norm of one image, use the binomial theorem.

> **Further hint**
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> After substituting the coherent-state series, replace the input number by the sum of the two output numbers. Simplify the factorials before separating the double series. Identify each normalised output factor before taking the partial trace.